SECTION 27 NOISE BARRIER WORKS

GENERAL

General 27.01 (1) The Noise Barrier Works (NBW) shall comprise all works above the concrete substructure of the noise barrier, including but not limited to main structural steel frame (such as vertical posts and cantilever beams), steel sub-frames (such as horizontal members), noise barrier panels and associated acoustical gasket profiles, clamping bars, acoustic sealing, fasteners, enclosure plates and sealants, fire breakers, access doors, base plates, holding down bolts, drainage gutters and downpipes, and any other parts and components necessary for the completion of the entire noise barrier as shown on the Drawings and herein specified.

(2) The following elements of the NBW are designed by the *Supervisor* (known as Conforming Design hereafter). Locations and details of which are shown on the Drawings.

(a) main structural steel frame, including holding down bolts to the substructures and connection details between the vertical posts and the base plate; and

(b) location of subframe of the noise barrier panels.

(3) All works in relation to the NBW including all details except those listed in sub-clause (2) above shall be designed by the *Contractor*'s Designer with reference to sub-clause (5) and (6) below as the *Contractor*'s Design in accordance with Clause F2 of *additional conditions of contract*. The *Contractor*'s Design of the NBW (known as NBW Design hereafter) shall include the following: -

(a) the sub-frames, the noise barrier panels and all the fixings details including the associated fittings, joints and flashing details. Such details, if any, as shown on the Drawings are indicative only;

(b) acoustical gasket profiles, clamping bars, acoustic sealing, fasteners, enclosure plates and sealants. Such details, if any, as shown on the Drawings are indicative only;

(c) connection and fixings details between the noise barriers panels and post of sign gantries; and

(d) any other parts or components necessary for the completion of the entire noise barrier system or as specified on the Drawings to be designed by the *Contractor*.

(4) The Noise Barrier Works Subcontractor (NBWSC) shall have proven experience and history in manufacturing and/or supplying noise barrier system, and shall satisfy the *Supervisor* by at least one project; the cost of the noise barriers for which shall exceed HK$0.5M; and was successfully completed in similar weathering conditions and to the codes, standards, and requirements in this Specification. The NBWSC should be accredited to ISO 9001.

(5) For the purpose of this Specification, the NBWSC shall be the manufacturer or agent of the noise barrier system. To ensure the integrity and continuous warranty of the soundness of the noise barrier, the NBWSC shall appoint a designer (known as *Contractor*'s Designer under this Specification) to carry out the NBW Design. The NBWSC shall execute the *works* in strict accordance with the requirements of the Contract. The NBWSC shall review the *Supervisor*'s Design in order to ensure the adequacy of the information as shown on the Drawings and compatibility with the NBW Design.

(6) As an alternative, the *Contractor* may carry out the review of the *Supervisor*'s Design himself, appoint a designer (known as *Contractor*'s Designer under this Specification) to carry out the NBW Design. If this is the case, the *Contractor* shall employ specialist workforce to install the noise barrier and appoint the NBSWC to supervise the NBW.

(7) During the review of the *Supervisor*'s Design by either the *Contractor* himself or by the NBWSC, if any of the details from the *Supervisor*'s Design are required to be modified to suit the NBW Design, the *Contractor* shall seek approval from the *Supervisor* and such request shall be made at least one month before commencement of the NBW Design.

(8) Structural steelwork shall be fabricated and installed by a structural steel specialist on the List of Approved Suppliers of Materials and Specialist Contractors for Public Works.

(9) The *Contractor* shall not engage more than one manufacturer or agent to undertake the NBW unless it is absolutely necessary, such as a single manufacturer for both absorptive type and reflective type is not available. In such case, the *Contractor* shall ensure the compatibility, the integrity, and the continuous warranty of soundness of the noise barrier manufactured by different manufacturers.

(10) The *Contractor*'s subcontracting of the Noise Barrier Works shall not relieve the *Contractor* from any liability or obligation under this contract and he shall be responsible for the acts, defaults and neglects of the Noise Barrier Works Subcontractor ("NBWSC") or the agents, employees or workers of the NBWSC as fully as if they were the acts, defaults or neglects of the *Contractor*.

(11)(a) Except in respect of those intellectual property rights referred to in sub-clause (c) of this Clause, the *Contractor* hereby undertakes and warrants to the *Client* that it is the sole legal and beneficial owner of all intellectual property rights subsisting in the design of the Noise Barrier Works

(b) Upon the issue of the certificate of Completion of whole of *the works* or after termination of the *Contractor*'s obligation to Provide the Works, or termination, abandonment or breach of contract, the *Contractor* shall be deemed to have granted to the *Client* and the subsequent owners or occupiers of the *works* free of all fees a transferable, non-exclusive and irrevocable licenses (carrying the right to grant sub-licences) to utilize the design of

the Noise Barrier Works in connection with execution of the *works* and/or the subsequent alteration, extension and maintenance thereof and for no other purpose whatsoever without the prior written agreement of the *Contractor*.

(c) To the extent that beneficial ownership of any intellectual property rights in the design of the Noise Barrier Works is vested in anyone other than the *Contractor*, the *Contractor* shall procure that the relevant beneficial owner shall grant a licence together with an indemnity to the *Client* and the subsequent owners or occupiers of the *works* upon the same terms mutatis mutandis as those set out in sub-clauses (11)(b) and (11)(f) of this Clause respectively.

(d) For the avoidance of doubt, any licence and indemnity granted pursuant to this Clause shall not be determined if the *Contractor* shall for any reason cease to be employed in connection with the works or the Contractor's obligation to Provide the Works to be terminated.

(e) The Contractor shall at the request of the *Client*, do such acts and execute all such deeds and documents (or procure that the same be done or executed) as the *Client* or the subsequent owners or occupiers of the works may require for vesting in the *Client* and the subsequent owners or occupiers of the works all or any of the rights referred to in this Clause. The *Contractor* shall bear its own costs and expenses in relation thereto.

(f) The Contractor hereby indemnifies the *Client* against all claims, proceedings, actions, damages and losses incurred or sustained by the *Client* arising from the use of the design of the Noise Barrier Works (irrespective of whether the intellectual property rights therein are owned by the *Contractor* or other parties) by the *Client* for purposes referred to in sub-clause (11) (b) of this Clause. For the avoidance of doubt, the indemnity herein applies where the proceedings concerned are subsequently withdrawn or settled or in the event that the allegations of infringement are subsequently found to be unsubstantiated. The *Contractor* shall at its own cost grant a like indemnity to the subsequent owners or occupiers of the *works* upon request of the *Client*.

Warranty 27.02 (1) The *Contractor* shall ensure that Noise Barrier Works shall remain in a condition conforming to the requirements set out in the Particular Specification for a period of 10 calendar years commencing from the date of Completion stated in the certificate of Completion of whole *of the works*.

(2) As part of the guarantee of the *Contractor* on the performance of the NBW, transparent noise barrier panels shall also carry a warranty on the properties specified in the Specification. The warranty shall be provided by NBWSC to the Government of the Hong Kong Special Administrative Region.

(3) The *Contractor* undertakes that in the event that the Noise Barrier Works fail to remain in the condition conforming to the requirements set out in this Section of Particular Specification for a period of 10 calendar years commencing from the date of Completion stated in the certificate of Completion of whole *of the*

works, issued as stated in sub-clause (1) of this Clause above, it shall as soon as reasonably practicable after receipt of notification by the *Client*, carry out at its own cost all repair or replacement work including emergency and temporary works which are necessary in the opinion of the *Client* to ensure that the Noise Barrier Works conform to the requirements set out in the Particular Specification. All repair or replacement work including emergency and temporary works shall be carried out by the *Contractor* at times as specified by the *Client* so as not to interfere with the proper use and functioning of the completed works. In the event that such repair or replacement work is not carried out by the *Contractor* in accordance with this sub-clause, the *Client* may, without prejudice to any other remedy, proceed to carry out such work by its own workers or by other contractors and the cost properly incurred by the *Client* in having such work carried out shall be a debt due to the *Client* from the *Contractor* and shall be paid to the *Client* by the *Contractor* immediately on demand.

(4) The *Contractor shall* enter into a sub-contract with the NBWSC in the Standard Form of Domestic Subcontract published by the Hong Kong Construction Association or such other form as may be approved by the *Project Manager* and shall within 21 days of so doing give to the *Client* a copy of the said executed sub-contract with only such amendments as may previously have been approved by the *Client* in writing.

(5) The *Contractor* shall be responsible for submitting to the *Client* within 21 days of the appointment of the NBWSC a Deed of Warranty duly executed by the NBWSC in favour of the *Client* in the form appearing at Appendix G42 with only such amendments as may previously have been approved by the *Client* in writing.

(6) The submission by the *Contractor* of the duly executed Deed of Warranty to the *Client* as referred to in sub-clause (5) of this Clause above is a condition precedent to the issue of the certificate of Completion.

MATERIALS

Design for Manufacturing and Assembly (DfMA) Technology

27.03 The *Contractor* is encouraged to adopt Design for Manufacturing and Assembly (DfMA) Technology for construction of NBW as far as possible to minimize wastage in transit from one site to another and facilitate re-use and use precast footprints to allow the re-cycled use.

Materials of noise barrier panels

27.04 All noise barrier panels shall be ultra-violet light resistance.

Alternative materials

27.05 The recommended materials used for the noise barrier panels shall be polymethyl methacrylate (PMMA). However this shall not preclude the need for the designer to identify as far as practicable other suitable materials for the transparent noise barrier panels and have all suitable materials specified on the Drawings for the *Contractor* to propose to the *Supervisor* for acceptance. Only in very special circumstances when there are no other suitable materials should the single material be specified. The *Contractor* shall not be entitled to compensation event

using alternative materials.

Transparent and absorptive noise barrier panels

27.06 (1) The panels shall be designed by the NBSWC but the structural elements, such as the associated fixing details of the panels shall be designed by the *Contractor*'s Designer to the requirements as stated in Clause 27.14.

(2) For quality assurance, transparent noise barrier panels are generally required to possess minimum five years of application history in weathering conditions similar to Hong Kong climate, and shall satisfy the *Supervisor* by at least one successful project for noise barriers on highways or similar works in Hong Kong in the past 3 years. In assessing the manufacturer or his agent's job references, project engineer may have his own set of considerations to suit individual case. As a general rule, a project involving at least $0.5 million worth of noise barriers with transparent panel works can be regarded as an acceptable job reference.

(3) The panels shall be of proprietary brand and produced by a manufacturer with ISO 9001 certification or equivalent. They shall bear the manufacturer's brush mark.

(4) The panels shall be supplied with dimensions to suit the framing arrangement as specified on the Drawings. Unless specially agreed by the *Supervisor*, alternative panel sizes requiring substantial changes in the framing or sub-framing shall not be allowed.

(5) The transparent noise barrier panels shall have a weighted sound reduction index of at least 25 dB which shall be tested in accordance with ZTV-Lsw88 or JIS 1416.

(6) The panels shall be ultra-violet light resistant.

PMMA panels

27.07 (1) (a) Polymethyl methacrylate (PMMA) noise barrier panels shall be either extruded or cast fully satisfying the ZTV-Lsw88 noise barrier standard and the requirements of this Specification.

(b) The PMMA noise barrier panels shall possess the following properties :

(i) light transmittance of at least

90 % for clear panel,

60 % for coloured panel,

50 % for translucent panel;

(ii) tensile strength of at least 65 N/mm^2;

(iii) modulus of elasticity of at least 2900 N/mm^2;

(iv) fire retardant property complying with the requirements in EN 1794-2 Annex A Class 2;

(v) stone impact resistance complying with the requirements in EN 1794-1 Annex C;

(vi) impact strength as Class A complying with BS 6206; and

(vii) no emission of noxious fumes or leachates as a result of fire.

(2) The NBWSC shall guarantee the panel properties at the end of the 10-year period, which shall include but not limited to the following :

(a) light transmittance loss not exceed 5% from the original value;

(b) tensile strength of at least 50 N/mm²;

(c) modulus of elasticity of at least 2600 N/mm²;

(d) colour difference shall be less than 20 units of ΔEab to DIN 6174;

(e) no mottling; and

(f) no loss in acoustic property, fire retardant property and stone impact resistance.

Polycarbonate panels

27.08 (1) (a) Polycarbonate noise barrier panels shall be coextruded (i.e. fusion), with weather protection on both sides against UV radiation from sunlight and for avoidance of coating failure. Unless specified otherwise in the Specification, requirements in ZTV-Lsw88 shall be fulfilled.

(b) The polycarbonate noise barrier panels shall possess the following properties

(i) light transmittance of at least

80 % for clear panel,

60 % for coloured panel,

50 % for translucent panel;

(ii) tensile strength of at least 55 N/mm²;

(iii) modulus of elasticity of at least 2100 N/mm²;

(iv) fire retardant property complying with the requirements in EN 1794-2 Annex A Class 2;

(v) stone impact resistance complying with the requirements in EN 1794-1 Annex C or JIS K 6735;

(vi) fire spread property complying with the requirements of Class 1 or Class 2 Surface Spread of Flame under

BS 476 Part 7; and

(vii) impact strength as Class A complying with BS 6206.

(2) The NBWSC shall guarantee the panel properties at the end of the 10-year period, which shall include but not limited to the following :

(a) light transmittance loss not exceed 7 % from the original value;

(b) tensile strength of at least 45 N/mm²;

(c) modulus of elasticity of at least 1800 N/mm²;

(d) yellowing index shall be less than 10 to ASTM D 1925 or JIS 7103;

(e) no mottling; and

(f) no loss in acoustic property, fire retardant property and stone impact resistance.

Sound absorptive panels

27.09 (1) The absorptive noise barriers shall be of both sides highly absorbing noise barrier element comprising, inter alia, front and rear perforated panels and dampening panels both sides with a minimum thickness of 120mm satisfying Sound Absorption Standard to DIN 52212,$\Delta L_{A,\alpha,Str}$: min 8dB, or approved products having equivalent functions or performance. The partition between the aluminum perforated panels shall comprise cement bonded wood chipboard, damping panels both sides from 40mm thick rockwool of 100kg/m^3, front which covered with black glass fleece, or other approved products having equivalent functions or performance. The absorptive coefficient of absorptive panels shall be at least 0.63.

(2) The absorptive panels shall achieve the same transmission loss requirements as transparent panels described in the Specification.

(3) The absorptive noise barriers shall be made of seawater resistant aluminum to DIN1725/1 with synthetic powder coating. The absorptive noise barrier panels shall comply with Noise Dampening Standard to DIN 52210, $\Delta L_{A,R,Str}$: min. 25dB and Noise Barrier Standard to ZTV-Lsw88.

(4) The pattern of the perforation on the panels shall be constituted by at least two types of hole and shall be submitted for approval by the *Supervisor*. Size of each perforated openings shall not exceed 5mm in diameter. The percentage of perforation shall be 30% minimum. The absorptive panels shall be non-welded, free draining, and free of pockets or cavities in which water may be collected. Panel shall be coated with a powder coating.

(5) The external colour of the absorptive noise barrier panels shall be as shown on the Drawings and subject to the agreement by the *Supervisor*.

(6) The absorptive noise barrier panels and the in-fill materials shall be weather resistant. Repeated cycles of soaking by water and subsequent drying shall not affect the acoustic performance and durability of the panels.

(7) The external and internal acoustical sealing located insides and outside of the absorptive noise barrier panels shall be extruded siliconized Ethylene Propylene Diena Monomer (EPDM) type gasket profile with hardness of 60 degree shore and with quality compatible with the noise barrier panels. Such gaskets shall be supplied by the noise barrier panel manufacturer and supported by an original test report.

(8) The NBWSC shall guarantee the panel properties at the end of 10-year period, which shall include but not limited to the following:

(a) Any acoustic fill material shall be non-corrosive, resistant to attack by fungus, fire-resistant, vermin proof, and non-hydroscopic. Fill materials shall also be water-resistant, free-draining, self-supporting and shall retain physical and sound absorptive characteristics after rain and long-term exposure to sunlight

(b) Free from any defects (materials and workmanship)

(c) No loss in acoustic property.

Supporting frames and mountings

27.10 (1) Conforming Design as defined in Clause 27.01(2) shall be in accordance with the Drawings. Any modifications required to suit the noise barrier panels shall be submitted to the *Supervisor* for approval.

(2) NBW Design as defined in Clause 27.01(3) are indicative only if they are shown on the Drawings.

(3) All steelworks of the noise barrier system shall comply with Section 18 of the Specification.

(4) Bolts connecting components of the structural frame and holding down bolts shall be stainless steel with strength grade A4-80 to the requirements as specified in Section 18 of the Specification.

(5) Structural steel shall be hot dip galvanized and then coated with System II of the painting system as specified in Section 18 of the Specification.

(6) The external and internal acoustical sealing located inside and outside of the noise barrier panels shall be extruded siliconized Ethylene Propylene Diena Monomer (EPDM) type gasket profile with hardness of 60 degree shore and with quality compatible with the noise barrier panel. Such gaskets shall be supplied by the NBWSC and supported by original test reports.

(7) Fixing of glass transparent noise barrier panels shall comply with the following requirements:

(a) Glass shall not be in direct contact with any other hard substance with hardness equivalent to or greater than the hardness of glass.

(b) For framed systems, the frame section shall give a minimum of 15 mm edge cover to the glass.

(c) For point bolted supports, tempered glass shall be used and the position of the connectors shall not allow the glass to undergo reverse curvature effects. Glass shall not come into direct contact with the bolts or clamping plates. There shall be clamping plates and gaskets on both sides of the glass that provide a minimum of 50 mm diameter cover to the glass. The cantilevered portion of the panels shall not be less than one-quarter of the span between the bolted connector.

(d) For clipped infill panels, the clips shall be positioned around the periphery of infill panels, at a maximum spacing of 600 mm. Each clip shall be at least 50 mm in length and shall give a minimum depth of cover to the glass of 25 mm.

(8) All fixings, fittings and any accessories shall be of fire retardant materials.

(9) Aluminium frames for transparent noise barrier shall be extruded from grade 6063-T6 aluminium with polyester powder coating. Colour shall be as shown in the Drawings and shall be to the satisfaction of the *Supervisor*.

SUBMISSIONS

Particulars of noise barrier works 27.11 (1) The *Contractor* shall submit to the *Supervisor* for approval the following particulars of the NBW at least 60 days prior to the commencement of the NBW :-

(a) Three copies of literature on the noise barrier panels from the panel manufacturer.

(b) Details of all components of the NBW, including any related test methods and results of the proposed materials.

(c) Three copies of full detailed design, drawings and calculations and one set of copy in both Microstation (.DGN) and Acrobat (.PDF) formats with Independent Checking Engineer‘s check certificate demonstrating the adequacy of all components of the NBW.

(d) Three copies of workshop drawings and one set of drawings in both Microstation (.DGN) and Acrobat (.PDF) formats showing the plan, sections and elevations of the NBW Works at a scale of not less than 1:100. The workshop drawings shall be certified by the Independent Checking Engineer. Exact location and spacing of the structural frames and all details necessary for the complete installation of the NBW shall be clearly shown.

(e) A certificate for each type of noise barrier panel showing the manufacturer's name, the date and place of manufacture and showing that the panels comply with the requirements stated in the Contract.

(f) Results of the following tests or equivalent to the satisfaction of the *Supervisor*:

(I) <u>for PMMA noise barrier panels</u>

(i) light transmittance to EN 1794-2 Annex F;

(ii) flexural strength to ISO 178;

(iii) impact strength complying with Class A to BS 6206;

(iv) stone impact resistance to EN 1794-1 Annex C;

(v) weighted sound reduction index to ZTV-Lsw88;

(vi) stability under wind load to EN 1794-1 Annex A;

(vii) fire retardant property to EN 1794-2 Annex A;

(viii) holding device to ZTV-Lsw88;

(ix) colour difference to DIN 6174; and

(x) catching contrivances, if applicable.

(II) <u>for polycarbonate noise barrier panels</u>

(i) light transmittance to EN 1794-2 Annex F;

(ii) tensile strength to ISO 527 or ASTM D 1003;

(iii) modulus of elasticity to ISO 527 or DIN 53457;

(iv) flexural strength to ISO 178;

(v) impact strength complying with Class A to BS 6206;

(vi) stone impact resistance to EN 1794-1 Annex C or JIS K 6735;

(vii) weighted sound reduction index to ZTV-Lsw88 or JIS 1416;

(viii) stability under wind load to EN 1794-1 Annex A;

(ix) holding device to ZTV-Lsw88;

(x) fire retardant property to EN 1794-2 Annex A;

(xi) fire spread property to BS 476 Part 7;

(xii) yellowing resistance to ASTM D 1925 or JIS 7103; and

(xiii) catching contrivances, if applicable.

(III) <u>for glass transparent noise barrier panels</u>

(i) light transmittance to EN 1794-2 Annex F;

(ii) tensile strength to ISO 527 or ASTM D 1003;

(iii) flexural strength to ISO 178;

(iv) impact strength complying with Class A to BS 6206;

(v) stone impact resistance to EN 1794-1 Annex C;

(vi) weighted sound reduction index to ZTV-Lsw88;

(vii) static load test to Clause 27.14(6);

(viii) holding device to ZTV-Lsw88;

(ix) fire spread property to BS 476 Part 7;

(x) colour difference to DIN 6174; and

(xi) catching contrivances, if applicable.

(g) Accelerated exposure test results in accordance with the guidelines stipulated in Appendix R, if applicable.

(h) Detailed proposal for Design for Manufacturing Assembly (DfMA) providing manufacture locations, DfMA components, method of installation, the associated temporary works, supervision and resource plan, quality assurance plan, etc.

(i) Programme of manufacture, testing and delivery, including name and address of testing institute.

(2) The *Contractor* shall be responsible for ensuring that sufficient time is allowed for submission and approval of his proposal as well as for ordering, production, testing and delivery of the noise barrier panels. Insufficient time for submission cannot substantiate itself in any case as a justification for applying alternative material.

(3) Approval of any calculations, drawings or method of installation by the *Supervisor* shall not relieve the *Contractor* of any of his

obligations under the Contract.

Samples 27.12 (1) The *Contractor* shall submit 3 sample panels of at least 0.5 m^2 for each type and colour of noise barrier panel to the *Supervisor* for approval of their finish, colour and quality at least 30 days prior to the ordering of noise barrier panels. Once approved by the *Supervisor*, these samples shall be retained as the quality standard throughout the Contract.

(2) The colour reference of each type of the noise barrier panels and of the structural steel members shown on the Drawings is subject to confirmation by the *Supervisor*. The colour reference shall be confirmed by the *Supervisor* as part of the approval of the sample panels and the *Contractor* shall obtain the confirmation prior to placing his order.

(3) The *Contractor* shall provide a full size mock up sample of each type of NBW. The mock up sample shall be at least 3 bays wide ("one bay" is defined as the panel length between noise barrier posts) and in full height or as directed by the *Supervisor*, and identical to the permanent NBW to be installed. The mock up sample shall be used to demonstrate workmanship. Location of the mock up sample shall be proposed by the *Contractor* for agreement by the *Supervisor*. The mock up sample shall not be incorporated into the permanent works unless otherwise agreed by the *Supervisor*.

As-constructed drawings and design calculations of noise barrier works 27.13 (1) The *Contractor* shall submit the as-constructed drawings (in which all information and details of the complete NBW installation as defined in Clause 27.01(1) shall be given) and design calculations of the *Contractor*'s Design of the NBW in accordance with Clause 1.80. In addition the contractor shall consult CHE/N (the maintenance office for highway structures) on the as-built records that should be submitted upon completion of the *works*.

(2) In addition to sub-clause (1) above, the *Contractor* shall also submit two sets of velographs within four weeks after substantial completion of the NBW to the *Supervisor*.

OPERATING CONDITIONS

Operating conditions 27.14 The noise barriers shall be designed to have a high level of aging and corrosion resistance to withstand the local climatic conditions specified hereunder: -

(a) The atmosphere under which the noise barriers will be operating is highly humid, salt laden and corrosive. The noise barriers shall therefore be tropicalized and vermin proofed. Corrosivity Category: C5-1 (Very High Industrial to BS EN 12944).

(b) The maximum and minimum operating temperature shall be taken as 42oC and 0oC respectively.

DESIGN REQUIREMENTS

Design requirements

27.15 (1) The NBW Design shall be designed to the requirements given in the latest version of "Structures Design Manual for Highways and Railways" published by Highways Department and "Guidelines on Design of Noise Barriers" jointly published by Highways Department and Department of Environmental Protection effective on the date before the closing date of the Tender. The dynamic pressure head to be adopted in the NBW Design shall be 3.3 kN/m².

(2) The noise barrier system shall incorporate holding devices to prevent panels from falling off the noise barrier system in the event of accidental impact by vehicle. Details of the holding devices shall be designed by the *Contractor*'s Designer for approval by the *Supervisor*. The holding devices shall comply with the following requirements:

(a) The devices shall hold the four corners of each panel member by wire ropes. Design holding force of the device shall be 10 kN.

(b) The devices shall allow a relative movement of 0.3 m between the panels and the structural frames.

Holding devices are deemed not necessary if external guard nets are provided in accordance with sub-clause (4) of this Clause in a very special circumstance.

(3) The noise barrier panels shall be fitted with catching contrivances or be provided with external guard nets made from wire mesh. Details of the guard nets shall be designed by the *Contractor*'s Designer for approval by the *Supervisor*, and catching contrivances shall either be:

(a) laminates between different sheets forming the panels; or

(b) embedded mesh within the panels.

(4) Those noise barriers vulnerable to direct vehicular impact and without catching contrivances, external guard nest made from wire mesh shall be provided. Details of the guard nets shall be designed by the *Contractor*'s Designer for approval by the *Supervisor*.

(5) Catching contrivances and external guard nets are deemed not necessary if the noise barrier is shown on the Drawings as erected on the concrete profile barrier, bridge parapet, and retaining wall, or if alternative measure is provided to the satisfaction of the *Supervisor*.

(6) To alleviate the potential problem with birds flying into transparent noise barriers, the transparent panels shall be equipped with:

(a) tinted materials, and

(b) embedded opaque stripes.

Incorporation of any of the above shall not adversely affect the overall appearance of the noise barrier and shall be subject to the satisfaction of the *Supervisor*. If the *Contractor* wishes to adopt the embedded opaque stripes or superimposed patterns of thin opaque stripes, he shall further substantiate to the satisfaction of the *Supervisor* that the stripes are durable and have no adverse effect on the performance and maintenance of the noise barrier panels.

(7) The noise barrier panels shall not reflect light in such a way as to prejudice road safety.

(8) The noise barrier shall be of the dimensions, geometry and the type as indicated on the Drawings or otherwise accepted by the *Supervisor*.

(9) Material used at overhead locations shall not form sharp and harmful pieces when shattered.

(10) In the case of noise barrier system whose elements whilst having the fire retardant property as specified in the Specification, are nevertheless combustible, the following provisions shall be incorporated into the system:

(a) a minimum 4 m wide section made of non-combustible material shall be required for every 100 m in length.

(b) panels on side wall shall be supported on raised plinth made of non-combustible material and shall be at least 800 mm above adjacent carriageway.

(c) The non-combustible section shall have the dimensions and locations as shown on the Drawings unless otherwise agreed with the *Supervisor*. The fire breakers shall be non-combustible and shall have the fire resistance property according to BS476-4:1970 or similar to be approved by the *Supervisor*.

(d) The color of non-combustible section shall match the colour scheme of adjacent noise barrier panel as shown on the Drawings unless otherwise agreed with the *Supervisor*.

(11) The stability of the noise barrier and its anchoring shall meet the requirements of EN 1794-1 Annex A under the wind loads as specified in sub-clause (1) of this Clause.

(12) Fixing details of the noise barrier shall be capable of spanning over the movement joint of the existing bridge structures or other structures and compatible with the maximum allowed movement range.

(13) Where panels are required to be curved this shall be done by thermoforming (hot bending) to the desired radius prior to delivery to site or otherwise as approved by the *Supervisor*.

(14) Should any street fire hydrants serving both bounds of carriageway be segregated by any noise barrier, a special panel which is hinged at one side and lockable to the noise barrier at the other side shall be designed by the *Contractor* and incorporated into the NBW. The size of the panel (or opening) shall be adequate to enable the connection of the fire hose to the fire hydrant from both bounds of the carriageway. In addition, the special panel shall bear an indication of the fire hydrant at the other side of the special panel, and equip with a pad lock. Details shall be subject to comments by FSD and approval by the *Supervisor*. An all-round clearance of 1.5m shall be maintained at each fire hydrant outlets and its controlling valve, unless otherwise agreed by the *Supervisor* after taking into account the comment by FSD.

TESTING

Testing 27.16 (1) All tests shall be carried out in accordance with internationally recognized standards and at an corresponding accredited testing institute proposed by the *Contractor* and approved by the *Supervisor*. Such tests shall have been carried out not more than 72 months for accelerated exposure test and 36 months for other tests, before the time of submitting the noise barrier proposal. All samples delivered to the testing institute shall be indelibly printed with the manufacturer‘s brushed mark for clear identification.

(2) If evidence cannot be produced that the proposed barrier panels have satisfactorily performed either locally or in similar weathering conditions for at least 5 years, the panel manufacturer shall carry out accelerated exposure test to substantiate the compliance of the panel properties with the specification requirements. The accelerated exposure test shall be carried out at an accredited testing institute proposed by the *Contractor* and approved by the *Supervisor* in accordance with the guidelines stipulated in Appendix R. Previous accelerated exposure test results could also be accepted provided the tests are properly performed and witnessed in a manner acceptable to the *Supervisor*.

(3) Upon delivery of the noise barrier panels on Site, the *Supervisor* shall at his discretion select samples from each type of noise barrier panel and order the tests as specified in Clause 27.11(1)(f) to be carried out on the selected samples as a quality control and continued monitoring of the barrier panels.

(4) If the result of any test stated in Clause 27.11(1)(f) does not comply with the specified requirements for the test, the test shall be carried out on additional samples to be selected by the *Supervisor*. The number of additional samples shall be twice the number for the original test. The noise barrier panel shall be considered as not complying with the specified requirements for the test if the result of any additional test does not comply with the specified requirements for the test.

(5) The sample panels shall be considered as complying with the

specified requirements for the accelerated exposure test stated in Clause 27.11(1)(g) if the two test specimens after the accelerated exposure test meet the following criteria :

(a) one specimen fully meets the specified requirements stipulated in Clause 27.07(2) for PMMA panels or Clause 27.08(2) for polycarbonate panels; and

(b) the other specimen at least meets 90 % of the specified requirements stipulated in Clause 27.07(2) for PMMA panels or Clause 27.08(2) for polycarbonate panels.

Otherwise, the sample panels shall be considered as not complying with the specified requirements for the test.

Each and every specimen referred to in this sub-clause shall have undergone the full exposure test including the exposure treatment stipulated in Appendix R and the physical tests.

(6) Static load test for glass transparent noise barrier panels shall comply with the following procedures and requirements:

(a) A representative prefabricated production sample unit, consisting of the glass panel, supporting frame and associated fixing components, shall be mounted in a test pressure chamber. The gap between chamber and glazing frame shall be clamped and sealed to withstand the pressure during the test without leakage.

(b) The test load shall be 1.25 times the design wind pressure of the noise barrier system and shall be maintained for at least 15 minutes.

(c) During the load test, the glass shall not break and the deflection of any structural element of the specimen shall not exceed 1/180 of the span or 20 mm, whichever is the lesser; and

(d) The extent of recovery of deformation 15 minutes after the removal of the test load shall be at least 95 % and the specimen shall not show any signs of separation, plastic deformation or deleterious effect.

(7) Static load test for PMMA and Polycarbonate noise barrier panels shall follow EN 1794-1 Annex A.

(8) The *Contractor* shall make adequate allowance for the quantity of sample panels required in the tests, including those used as control samples, in his order of noise barrier panels and for the additional time required for the tests to be carried out on the sample panels in his programme.

INSTALLATION

Installation 27.17 (1) The NBW shall be designed and constructed using Design for Manufacturing and Assembly (DfMA). The NBW including

steelwork main frame, sub-frame and accessories.

(2) The NBW shall be installed to the details approved by the *Supervisor*. Provisions shall be made for both the horizontal and vertical adjustments of the structural frame and the noise barrier panels during installation.

(3) Site storage of materials shall comply with the recommendations and instructions of the NBWSC.

(4) After installation, framing members and accessories shall be cleaned by an approved method. Transparent noise barrier panels shall be left in clear and scratch free condition, both inside and outside, with all labels removed. Abrasive materials of any kind and sharp instruments shall not be used in cleaning surfaces. Benzene, leaded gasoline, acetone, carbon tetrachloride and cleaning liquids containing abrasives or powder cleaners shall not be used in cleaning surfaces. All rubbish, cartons crates and any excess materials from the manufacture, delivery and installation shall be removed.

INSPECTION AND MAINTENANCE

Inspection and maintenance manual

27.18 (1) Any parts and components of the NBW shall be designed to be inspectable, maintainable, and replaceable without causing damage to other parts or components.

(2) An Inspection and Maintenance Manual (IMM) of the NBW shall be submitted for approval by the *Supervisor*. The IMM shall include but not limited to any technical and as-constructed details of the NBW incorporated into the *works* and the recommended inspection and maintenance requirements, such as the type, frequency, and the procedure for undertaking inspection and maintenance. Format of the IMM shall make reference to the ”Inspection Manual for Highway Structures” and ”Road Inspection Manual” published by Highways Department and shall be agreed with the *Supervisor*. The *Contractor* shall submit draft IMM for comments by the *Supervisor* and the maintenance authorities. The *Contractor* shall satisfactorily address all comments and revise the Manual before submitting the final version.

SPARE PARTS

Provision of spare parts

27.19 (1) The *Contractor* shall supply, deliver and handover below spare parts to the maintenance authorities.

(a) 10 nos PMMA panel sized 2000mm(H) x 3000mm(W) x 20mm(THK)

(b) 10 nos Polycarbonate panel sized 1000mm(H) x 3000mm(W) x 20mm(THK)

(c) 100 nos M33 stainless steel hexagon nuts with washers

The spare part materials shall comply with the requirements stated in this Specification and Drawings of the Contract.

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